



Office of Academic Programs
2628 University Hall (MC 103)
601 South Morgan Street
Chicago, Illinois 60607-7126

February 5, 2020

TO: Tony Corte, Chair
Senate Committee on Educational Policy

FROM: Dana Wright, Director of Academic Program Development

I am submitting for the review and action of the Senate Committee on Educational Policy the attached **Proposal to Revise the Bachelor of Science in Chemical Engineering**.

The proposal and affiliated courses were approved by the Department of Chemical Engineering (multiple dates in 2019) and the College of Engineering's Educational Policy Committee (November 18, 2019).

ATTACHMENT

<i>Title:</i>	Revision of the Bachelor of Science in Chemical Engineering
<i>Sponsor:</i>	<i>Department of Chemical Engineering</i>
<i>Executive Summary:</i>	The department of chemical engineering is proposing to modify its BS-Degree requirements by: - Introducing six new courses: - CHE 230: Molecular Systems in Chemical Engineering - CHE 330: Polymer Science - CHE 453: Fundamentals of Electrochemistry - CHE 454: Molecular and Macromolecular Engineering - CHE 455: Nanoscale Systems in Chemical Engineering - CHE 457: Colloidal and Interfacial Phenomena - Adding an option to choose between CHE 330 Polymer Science or CHEM 346 Physical Chemistry II as a required course for the Bachelor of Science in Chemical Engineering degree. - Removing the following 400-level courses: - CHE 410 (Transport Phenomena) - CHE 431 (Numerical Methods in Chemical Engineering) - CHE 445 (Mathematical Methods in Chemical Engineering)
<i>Description:</i>	Add two courses, CHE 230 (which will be used in the development of a new concentration to be submitted in a separate proposal), CHE 330 (to be used to fulfil a selective requirement), and four technical electives, CHE 453, CHE 454, CHE 455, CHE 457, inside the major rubric. Add the following option; Choose CHE 330 or CHEM 346 inside the major rubric. Remove three technical electives CHE 410, 431, 445 inside the major rubric. A separate revision to the MS and PhD in Chemical Engineering proposes converting CHE 410, 431, 445 to graduate (500-level courses). The changes do not affect the total credit hours required for the program. Note: The revision of the MS and PhD can be made administratively by the Office of Academic Program Development as the only changes to the courses are their number.
<i>Justification:</i>	The proposed courses CHE 230, 330 will introduce several topics that are important to traditional and emerging fields in chemical engineering; polymers, molecular engineering, macromolecular systems, and nanomaterials. The proposed courses; CHE 453, 454, 455 will provide more comprehensive knowledge of the above topics. CHE 457 will be added as a technical elective to provide knowledge of colloids and interfacial phenomena. These topics have been taught previously as Selected Topics in Chemical Engineering (CHE 494 courses), but not as permanent courses

	in our curriculum. The addition of these 5 courses will strengthen our core curriculum in areas key to modern chemical engineering practice where they play a central role in biological, chemical/industrial, environmental and pharmaceutical processes and technologies. Ultimately, they will also provide the required curriculum for 5 new chemical engineering concentrations in development that will be proposed to our core program; Polymer and Molecular Engineering, Nanotechnology, Energy and Environment, Process Automation, and Intellectual Property. 2. Polymer science is an important topic that is currently not being covered in our curriculum. Most of our students work in industry where the material taught in CHE 330 will be very useful. In order to allow students to take CHE 330 Polymer Science without increasing the total credit hours in our program, students will be given the option to choose between CHE 330 or CHEM 346. CHEM 346 Physical Chemistry II, an advanced chemistry course, is currently required in our core program but it is not needed to satisfy the 30 hour ABET Math/Science requirement. The chemical engineering curriculum has a total of 52 credit hours of Math/Science courses, with 30 hours of chemistry courses. We have met with the Chemistry Department to inform them of our proposed change and took their suggestion to offer CHE 330 Fall semester so that chemical engineering students may still take CHEM 346 when it is offered in the Spring semester. 3. CHE 410, 431 and 445 are advanced chemical engineering courses originally created at the 400-level to allow undergraduates to take as the chemical engineering technical elective. The courses are taught at the graduate level and for the past 5 years only graduate students have enrolled. Changing CHE 410, 431 and 445 to CHE 520, 531 and 545 will not affect the undergraduate program; there are many other technical elective courses for the undergraduates to choose from for their technical elective requirement.
<i>Catalog Statement:</i>	<i>See below</i>
<i>Minority Impact Statement:</i>	<i>The changes will clarify and improve the degree requirements for all students and will not adversely impact minority students.</i>
<i>Budgetary and Staff Implications:</i>	<i>None expected</i>
<i>Library Resource Implications:</i>	<i>None expected</i>
<i>Space Implications:</i>	<i>None expected</i>

<i>Unit (e.g. department) approval date:</i> <i>College (educational policy committee, faculty) approval dates:</i>	Provide unit dates, including department, college/school committees, faculty Approvals, as applicable. Chemical engineering faculty approval: 05/13/19 faculty meeting Chemical Engineering department approval: 10/09/19 for CHE 230, 330, 454, 455, 10/16/19 for CHE 410, 431 and 445, and 11/05/19 for CHE 457 Education Policy Committee approval: 11/8/19
<i>Contact Person:</i>	Alan Zdunek zdunek@uic.edu
<i>Proposed Effective Date/Term:</i>	Fall 2020

Current	Revised																																																																						
BS in Chemical Engineering <i>Degree Requirements</i> To earn a Bachelor of Science in Chemical Engineering degree from UIC, students need to complete University, college, and department degree requirements. The Department of Chemical Engineering degree requirements are outlined below. Students should consult the <i>College of Engineering</i> section for additional degree requirements and college academic policies. BS in Chemical Engineering Degree Requirements Hours <table> <tr> <td>Nonengineering and General Education Requirements</td><td>73</td></tr> <tr> <td>Required in the College of Engineering</td><td>49</td></tr> <tr> <td>Technical Elective^a</td><td>3</td></tr> <tr> <td>Electives outside the Major Rubric^a</td><td>3</td></tr> <tr> <td>Total Hours—BS in Chemical Engineering</td><td>128</td></tr> </table> <i>a Students in the Biochemical Engineering Concentration take a minimum of 8 hours of electives and 130 hours for the degree; see below.</i> Nonengineering and General Education Requirements <table> <tr> <td>ENGL 160— Academic Writing I: Writing for Academic and Public Contexts</td><td>3</td></tr> <tr> <td>ENGL 161— Academic Writing II: Writing for Inquiry and Research</td><td>3</td></tr> <tr> <td>Exploring World Cultures course^a</td><td>3</td></tr> <tr> <td>Understanding the Creative Arts course^a</td><td>3</td></tr> <tr> <td>Understanding the Past course^a</td><td>3</td></tr> <tr> <td>Understanding the Individual and Society course^a</td><td>3</td></tr> <tr> <td>Understanding U.S. Society course^a</td><td>3</td></tr> <tr> <td>MATH 180—Calculus I^b</td><td>4</td></tr> <tr> <td>MATH 181—Calculus II^b</td><td>4</td></tr> <tr> <td>MATH 210—Calculus III^b</td><td>3</td></tr> <tr> <td>MATH 220—Introduction to Differential Equations</td><td>3</td></tr> <tr> <td>PHYS 141—General Physics I (Mechanics)^b</td><td>4</td></tr> </table>	Nonengineering and General Education Requirements	73	Required in the College of Engineering	49	Technical Elective ^a	3	Electives outside the Major Rubric ^a	3	Total Hours—BS in Chemical Engineering	128	ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3	ENGL 161— Academic Writing II: Writing for Inquiry and Research	3	Exploring World Cultures course ^a	3	Understanding the Creative Arts course ^a	3	Understanding the Past course ^a	3	Understanding the Individual and Society course ^a	3	Understanding U.S. Society course ^a	3	MATH 180—Calculus I ^b	4	MATH 181—Calculus II ^b	4	MATH 210—Calculus III ^b	3	MATH 220—Introduction to Differential Equations	3	PHYS 141—General Physics I (Mechanics) ^b	4	BS in Chemical Engineering SAME BS in Chemical Engineering Degree Requirements Hours <table> <tr> <td>Nonengineering and General Education Requirements</td><td>70</td></tr> <tr> <td>Required in the College of Engineering</td><td>49</td></tr> <tr> <td>Technical Elective^a</td><td>3</td></tr> <tr> <td>Selective</td><td>3</td></tr> <tr> <td>Electives outside the Major Rubric^a</td><td>3</td></tr> <tr> <td>Total Hours—BS in Chemical Engineering</td><td>128</td></tr> </table> Nonengineering and General Education Requirements <table> <tr> <td>ENGL 160— Academic Writing I: Writing for Academic and Public Contexts</td><td>3</td></tr> <tr> <td>ENGL 161— Academic Writing II: Writing for Inquiry and Research</td><td>3</td></tr> <tr> <td>Exploring World Cultures course^a</td><td>3</td></tr> <tr> <td>Understanding the Creative Arts course^a</td><td>3</td></tr> <tr> <td>Understanding the Past course^a</td><td>3</td></tr> <tr> <td>Understanding the Individual and Society course^a</td><td>3</td></tr> <tr> <td>Understanding U.S. Society course^a</td><td>3</td></tr> <tr> <td>MATH 180—Calculus I^b</td><td>4</td></tr> <tr> <td>MATH 181—Calculus II^b</td><td>4</td></tr> <tr> <td>MATH 210—Calculus III^b</td><td>3</td></tr> <tr> <td>MATH 220—Introduction to Differential Equations</td><td>3</td></tr> <tr> <td>PHYS 141—General Physics I (Mechanics)^b</td><td>4</td></tr> </table>	Nonengineering and General Education Requirements	70	Required in the College of Engineering	49	Technical Elective ^a	3	Selective	3	Electives outside the Major Rubric ^a	3	Total Hours—BS in Chemical Engineering	128	ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3	ENGL 161— Academic Writing II: Writing for Inquiry and Research	3	Exploring World Cultures course ^a	3	Understanding the Creative Arts course ^a	3	Understanding the Past course ^a	3	Understanding the Individual and Society course ^a	3	Understanding U.S. Society course ^a	3	MATH 180—Calculus I ^b	4	MATH 181—Calculus II ^b	4	MATH 210—Calculus III ^b	3	MATH 220—Introduction to Differential Equations	3	PHYS 141—General Physics I (Mechanics) ^b	4
Nonengineering and General Education Requirements	73																																																																						
Required in the College of Engineering	49																																																																						
Technical Elective ^a	3																																																																						
Electives outside the Major Rubric ^a	3																																																																						
Total Hours—BS in Chemical Engineering	128																																																																						
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3																																																																						
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3																																																																						
Exploring World Cultures course ^a	3																																																																						
Understanding the Creative Arts course ^a	3																																																																						
Understanding the Past course ^a	3																																																																						
Understanding the Individual and Society course ^a	3																																																																						
Understanding U.S. Society course ^a	3																																																																						
MATH 180—Calculus I ^b	4																																																																						
MATH 181—Calculus II ^b	4																																																																						
MATH 210—Calculus III ^b	3																																																																						
MATH 220—Introduction to Differential Equations	3																																																																						
PHYS 141—General Physics I (Mechanics) ^b	4																																																																						
Nonengineering and General Education Requirements	70																																																																						
Required in the College of Engineering	49																																																																						
Technical Elective ^a	3																																																																						
Selective	3																																																																						
Electives outside the Major Rubric ^a	3																																																																						
Total Hours—BS in Chemical Engineering	128																																																																						
ENGL 160— Academic Writing I: Writing for Academic and Public Contexts	3																																																																						
ENGL 161— Academic Writing II: Writing for Inquiry and Research	3																																																																						
Exploring World Cultures course ^a	3																																																																						
Understanding the Creative Arts course ^a	3																																																																						
Understanding the Past course ^a	3																																																																						
Understanding the Individual and Society course ^a	3																																																																						
Understanding U.S. Society course ^a	3																																																																						
MATH 180—Calculus I ^b	4																																																																						
MATH 181—Calculus II ^b	4																																																																						
MATH 210—Calculus III ^b	3																																																																						
MATH 220—Introduction to Differential Equations	3																																																																						
PHYS 141—General Physics I (Mechanics) ^b	4																																																																						

PHYS 142—General Physics II (Electricity and Magnetism) ^b	4	PHYS 142—General Physics II (Electricity and Magnetism) ^b	4
CHEM 122- General College Chemistry I ^b (Lecture)	4	CHEM 122- General College Chemistry I ^b (Lecture)	4
CHEM 123- General College Chemistry I ^b (Laboratory)	1	CHEM 123- General College Chemistry I ^b (Laboratory)	1
<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5	<u>OR</u> CHEM 116 – Honors and Major General and Analytical Chemistry I	5
CHEM 124—General College Chemistry II ^b (Lecture)	4	CHEM 124—General College Chemistry II ^b (Lecture)	4
CHEM 125—General College Chemistry II ^b (Lab)	1	CHEM 125—General College Chemistry II ^b (Lab)	1
<u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II	5	<u>OR</u> CHEM 118 – Honors and Major General and Analytical Chemistry II	5
CHEM 222—Analytical Chemistry ^c	4	CHEM 222—Analytical Chemistry ^c	4
CHEM 232—Organic Chemistry I	4	CHEM 232—Organic Chemistry I	4
CHEM 233- Organic Chemistry Laboratory	2	CHEM 233- Organic Chemistry Laboratory	2
CHEM 234—Organic Chemistry II	4	CHEM 234—Organic Chemistry II	4
CHEM 342—Physical Chemistry I	3	CHEM 342—Physical Chemistry I	3
CHEM 346—Physical Chemistry II	3	CHEM 346	
Total Hours- Nonengineering and General Education Requirements	73	Total Hours- Nonengineering and General Education Requirements	70
a Students should consult the General Education section of the catalog for a list of approved courses in this category.		SAME	
b This course is approved for the Analyzing the Natural World General Education category.			
^c Students who take CHEM 116 and CHEM 118 to fulfill the general chemistry requirement do not need to take CHEM 222. Instead they should enroll in one of the following: CHEM 314: Inorganic Chemistry; CHEM 452: Biochemistry; or CHEM 402: Chemical Information Systems and CHEM 444: Physical Chemistry III.		SAME	
Required in the College of Engineering		SAME	
Courses	Hours		
ENGR 100—Orientation ^a	0 ^a		
CHE 201—Introduction to Thermodynamics	3		
CHE 205—Computational Methods in Chem. Eng.	3		
CHE 210—Material and Energy Balances	4		
CHE 301—Chemical Engineering: Thermodynamics	3		
CHE 311—Transport Phenomena I	3		
CHE 312—Transport Phenomena II	3		
CHE 313—Transport Phenomena III	3		
CHE 321—Chemical Reaction Engineering	3		
CHE 341—Chemical Process Control	3		
CHE 381—Chemical Engineering Laboratory I	2		
CHE 382—Chemical Engineering Laboratory II	2		
CHE 396—Senior Design I	4		
CHE 397—Senior Design II	4		
CHE 499—Professional Development Seminar	0		
ECE 210—Electrical Circuit Analysis	3		
CME 260—Properties of Materials	3		
CS 109—C/C++ Programming for Engineers with MatLab	3		
Total Hours—Required in the College of Engineering	49		

a ENGR 100 is one-semester-hour course, but the hour does not count toward the total hours required for graduation.	SAME Selective: Choose from CHE 330 and CHEM 346 (3) <i>Note for Degree Audit: Selective will be used in calculating students' major GPA.</i> Total Hours: Selective 3																																																																																								
Technical Elective <table border="0"> <thead> <tr> <th>Courses</th><th>Hours</th></tr> </thead> <tbody> <tr> <td><i>One technical elective to be chosen from the following list of design-oriented courses:^a</i></td><td>3</td></tr> <tr> <td>CHE 392—Undergraduate Research (3)^b</td><td></td></tr> <tr> <td>CHE 410—Transport Phenomena (3)</td><td></td></tr> <tr> <td>CHE 413—Introduction to Flow in Porous Media (3)</td><td></td></tr> <tr> <td>CHE 421—Combustion Engineering (3)</td><td></td></tr> <tr> <td>CHE 422—Biochemical Engineering (3)</td><td></td></tr> <tr> <td>CHE 423—Catalytic Reaction Engineering (3)</td><td></td></tr> <tr> <td>CHE 425 Nanotechnology Drug Deliver (3)</td><td></td></tr> <tr> <td>CHE 431—Numerical Methods in Chemical Engineering (3)</td><td></td></tr> <tr> <td>CHE 433 Process Simulation with Aspen Plus (3)</td><td></td></tr> <tr> <td>CHE 438—Computational Molecular Modeling (3)</td><td></td></tr> <tr> <td>CHE 440—Non-Newtonian Fluids (3)</td><td></td></tr> <tr> <td>CHE 441—Computer Applications in Chemical Engineering (3)</td><td></td></tr> <tr> <td>CHE 445—Mathematical Methods in Chemical Engineering (3)</td><td></td></tr> <tr> <td>CHE 450—Air Pollution Engineering (4)</td><td></td></tr> <tr> <td>CHE 451 Renewable Energy Technologies (3)</td><td></td></tr> <tr> <td>CHE 456—Fundamentals and Design of Microelectronics Processes 930</td><td></td></tr> <tr> <td>CHE 494—Selected Topics in Chemical Engineering (3)</td><td></td></tr> <tr> <td>Total Hours—Technical Elective</td><td>3</td></tr> </tbody> </table> a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering. ^cCurrent <u>CHE 494</u> courses offered include: <i>Electrochemistry, Entrepreneurship in Chemical Engineering, 2D Nanomaterials, Membranes, Nanotechnology and Bio-Nanotechnology, Polymer Materials</i>	Courses	Hours	<i>One technical elective to be chosen from the following list of design-oriented courses:^a</i>	3	CHE 392—Undergraduate Research (3) ^b		CHE 410—Transport Phenomena (3)		CHE 413—Introduction to Flow in Porous Media (3)		CHE 421—Combustion Engineering (3)		CHE 422—Biochemical Engineering (3)		CHE 423—Catalytic Reaction Engineering (3)		CHE 425 Nanotechnology Drug Deliver (3)		CHE 431—Numerical Methods in Chemical Engineering (3)		CHE 433 Process Simulation with Aspen Plus (3)		CHE 438—Computational Molecular Modeling (3)		CHE 440—Non-Newtonian Fluids (3)		CHE 441—Computer Applications in Chemical Engineering (3)		CHE 445—Mathematical Methods in Chemical Engineering (3)		CHE 450—Air Pollution Engineering (4)		CHE 451 Renewable Energy Technologies (3)		CHE 456—Fundamentals and Design of Microelectronics Processes 930		CHE 494—Selected Topics in Chemical Engineering (3)		Total Hours—Technical Elective	3	<table border="0"> <thead> <tr> <th>Courses</th><th>Hours</th></tr> </thead> <tbody> <tr> <td><i>One technical elective to be chosen from the following list of design-oriented courses:^a</i></td><td>3</td></tr> <tr> <td>CHE 392—Undergraduate Research (3)^b</td><td></td></tr> <tr> <td>Delete 410</td><td></td></tr> <tr> <td>CHE 413—Introduction to Flow in Porous Media (3)</td><td></td></tr> <tr> <td>CHE 421—Combustion Engineering (3)</td><td></td></tr> <tr> <td>CHE 422—Biochemical Engineering (3)</td><td></td></tr> <tr> <td>CHE 423—Catalytic Reaction Engineering (3)</td><td></td></tr> <tr> <td>CHE 425 Nanotechnology Drug Deliver (3)</td><td></td></tr> <tr> <td>Delete CHE 431</td><td></td></tr> <tr> <td>CHE 433 Process Simulation with Aspen Plus (3)</td><td></td></tr> <tr> <td>CHE 438—Computational Molecular Modeling (3)</td><td></td></tr> <tr> <td>CHE 440—Non-Newtonian Fluids (3)</td><td></td></tr> <tr> <td>CHE 441—Computer Applications in Chemical Engineering (3)</td><td></td></tr> <tr> <td>Delete CHE 445</td><td></td></tr> <tr> <td>CHE 450—Air Pollution Engineering (4)</td><td></td></tr> <tr> <td>CHE 451 Renewable Energy Technologies (3)</td><td></td></tr> <tr> <td>CHE 453- Fundamentals of Electrochemistry (3)</td><td></td></tr> <tr> <td>CHE 454 Molecular and Macromolecular Engineering (3)</td><td></td></tr> <tr> <td>CHE 455 Nanoscale Systems in Chemical Engineering (3)</td><td></td></tr> <tr> <td>CHE 456—Fundamentals and Design of Microelectronics Processes (3)</td><td></td></tr> <tr> <td>CHE 457 Colloids and Interfacial Phenomena (3)</td><td></td></tr> <tr> <td>CHE 494—Selected Topics in Chemical Engineering (3)</td><td></td></tr> <tr> <td>Total Hours—Technical Elective</td><td>3</td></tr> </tbody> </table> a Possible technical elective credit for a 400-level CHE course not listed above will require departmental approval by petition to the Undergraduate Committee. b An appropriate design-related research project may be selected with the approval of the Department of Chemical Engineering. REMOVE FOOTNOTE “C”	Courses	Hours	<i>One technical elective to be chosen from the following list of design-oriented courses:^a</i>	3	CHE 392—Undergraduate Research (3) ^b		Delete 410		CHE 413—Introduction to Flow in Porous Media (3)		CHE 421—Combustion Engineering (3)		CHE 422—Biochemical Engineering (3)		CHE 423—Catalytic Reaction Engineering (3)		CHE 425 Nanotechnology Drug Deliver (3)		Delete CHE 431		CHE 433 Process Simulation with Aspen Plus (3)		CHE 438—Computational Molecular Modeling (3)		CHE 440—Non-Newtonian Fluids (3)		CHE 441—Computer Applications in Chemical Engineering (3)		Delete CHE 445		CHE 450—Air Pollution Engineering (4)		CHE 451 Renewable Energy Technologies (3)		CHE 453- Fundamentals of Electrochemistry (3)		CHE 454 Molecular and Macromolecular Engineering (3)		CHE 455 Nanoscale Systems in Chemical Engineering (3)		CHE 456—Fundamentals and Design of Microelectronics Processes (3)		CHE 457 Colloids and Interfacial Phenomena (3)		CHE 494—Selected Topics in Chemical Engineering (3)		Total Hours—Technical Elective	3
Courses	Hours																																																																																								
<i>One technical elective to be chosen from the following list of design-oriented courses:^a</i>	3																																																																																								
CHE 392—Undergraduate Research (3) ^b																																																																																									
CHE 410—Transport Phenomena (3)																																																																																									
CHE 413—Introduction to Flow in Porous Media (3)																																																																																									
CHE 421—Combustion Engineering (3)																																																																																									
CHE 422—Biochemical Engineering (3)																																																																																									
CHE 423—Catalytic Reaction Engineering (3)																																																																																									
CHE 425 Nanotechnology Drug Deliver (3)																																																																																									
CHE 431—Numerical Methods in Chemical Engineering (3)																																																																																									
CHE 433 Process Simulation with Aspen Plus (3)																																																																																									
CHE 438—Computational Molecular Modeling (3)																																																																																									
CHE 440—Non-Newtonian Fluids (3)																																																																																									
CHE 441—Computer Applications in Chemical Engineering (3)																																																																																									
CHE 445—Mathematical Methods in Chemical Engineering (3)																																																																																									
CHE 450—Air Pollution Engineering (4)																																																																																									
CHE 451 Renewable Energy Technologies (3)																																																																																									
CHE 456—Fundamentals and Design of Microelectronics Processes 930																																																																																									
CHE 494—Selected Topics in Chemical Engineering (3)																																																																																									
Total Hours—Technical Elective	3																																																																																								
Courses	Hours																																																																																								
<i>One technical elective to be chosen from the following list of design-oriented courses:^a</i>	3																																																																																								
CHE 392—Undergraduate Research (3) ^b																																																																																									
Delete 410																																																																																									
CHE 413—Introduction to Flow in Porous Media (3)																																																																																									
CHE 421—Combustion Engineering (3)																																																																																									
CHE 422—Biochemical Engineering (3)																																																																																									
CHE 423—Catalytic Reaction Engineering (3)																																																																																									
CHE 425 Nanotechnology Drug Deliver (3)																																																																																									
Delete CHE 431																																																																																									
CHE 433 Process Simulation with Aspen Plus (3)																																																																																									
CHE 438—Computational Molecular Modeling (3)																																																																																									
CHE 440—Non-Newtonian Fluids (3)																																																																																									
CHE 441—Computer Applications in Chemical Engineering (3)																																																																																									
Delete CHE 445																																																																																									
CHE 450—Air Pollution Engineering (4)																																																																																									
CHE 451 Renewable Energy Technologies (3)																																																																																									
CHE 453- Fundamentals of Electrochemistry (3)																																																																																									
CHE 454 Molecular and Macromolecular Engineering (3)																																																																																									
CHE 455 Nanoscale Systems in Chemical Engineering (3)																																																																																									
CHE 456—Fundamentals and Design of Microelectronics Processes (3)																																																																																									
CHE 457 Colloids and Interfacial Phenomena (3)																																																																																									
CHE 494—Selected Topics in Chemical Engineering (3)																																																																																									
Total Hours—Technical Elective	3																																																																																								

Courses	Hours	SAME
Electives outside the CHE rubric	3	
Total Hours—Electives outside the Major Rubric	3	
BS in Chemical Engineering, Bioengineering Concentration		SAME
Students in this concentration complete the following:		
Required Courses ^a		
Technical Elective		
<u>CHE 422</u> Biochemical Engineering	3	
Electives		
Select two electives in nonmajor rubric category from the following:	5-7	
<u>BIOS 350</u> General Microbiology		
<u>BIOS 351</u> Microbiology Laboratory		
<u>CHEM 352</u> Introductory Biochemistry		
<u>CHEM 452</u> Biochemistry I		
Total Hours	8-10	
^a Due to structure of the concentration and the prerequisites required for some of the courses, students in the concentration will be required to take a minimum of 130 semester hours for the degree.		
Sample Course Schedule		
Freshman Year		
First Semester	Hours	
MATH 180—Calculus I	4	
CHEM 122—General College Chemistry I (Lecture)	4	
CHEM 123—General College Chemistry I (Lab)	4	1
OR CHEM 116 – Honors and Major General and Analytical Chemistry I	5	
ENGL 160—Academic Writing I: Writing for Academic and Public Contexts	3	
CS 109—C/C++ Programming for Engineers with MatLab	3	
ENGR 100—Orientation ^a	1 ^a	
Total Hours	15	
Second Semester		
MATH 181—Calculus II	4	
PHYS 141—General Physics I (Mechanics)	4	
ENGL 161—Academic Writing II: Writing for Inquiry and Research	3	
CHEM 124—General College Chemistry II (Lecture)	4	
		SAME

CHEM 125—General College Chemistry II (Lab)	1		
OR CHEM 118— Honors and Major General and Analytical Chemistry II	5	SAME	
Total Hours	16		
Sophomore Year			
First Semester	Hours		
MATH 210—Calculus III	3		
PHYS 142—General Physics II (Electricity and Magnetism)	4	SAME	
CHEM 232—Organic Chemistry I	4		
CHE 201—Introduction to Thermodynamics	3		
General Education Core course	3		
Total Hours	17		
Second Semester	Hours		
MATH 220—Introduction to Differential Equations	3		
CHEM 234—Organic Chemistry II	4		
CHEM 233—Organic Chemistry Lab I	2		
		SAME	
CHE 205—Computational Methods in Chem. Eng.	3		
CHE 210—Material and Energy Balances	4		
CME 260— Properties of Materials	3		
Total Hours	19		
Junior Year			
First Semester	Hours		
ECE 210—Electrical Circuit Analysis	3	SAME	
CHE 301—ChE Thermodynamics	3		
CHE 311—Transport Phenomena I	3		
CHEM 222—Analytical Chemistry	4		
General Education Core course	3		
Total Hours	16		
Second Semester	Hours		
CHEM 342—Physical Chemistry I	3		
CHE 312—Transport Phenomena II	3		
CHE 313— Transport Phenomena III	3	SAME	
CHE 321—Chemical Reaction Engineering	3		
General Education Core course	3		
Total Hours	15		
Senior Year			
First Semester	Hours		
CHE 381—Chemical Engineering Laboratory I	2		
CHE 396—Senior Design I	4		
CHEM 346—Physical Chemistry II	3		
CHE Technical elective—Selected from CHE 410, 413, 421, 422, 423, 431, 438, 440, 441, 445, 450, 456, 494, or 392 (departmental approval is required for CHE 392)	3		
General Education Core Course	3		
Total Hours	15		
		Senior Year	
		First Semester	Hours
		CHE 381—Chemical Engineering Laboratory I	2
		CHE 396—Senior Design I	4
		CHE 330 – Polynmer Science or CHEM 346—Physical Chemistry II	3
		CHE Technical elective—Selected from list of courses above	3
		General Education Core Course	3
		Total Hours	15

Second Semester	Hours	Same
CHE 382—Chemical Engineering Laboratory II	2	
CHE 341—Chemical Process Control	3	
CHE 397—Senior Design II	4	
CHE 499—Professional Development Seminar	0	
Elective outside the major rubric	3	
General Education Core course	3	
Total Hours	15	